TurtleBot Autonomous Navigation with a Custom RRT* Global Planner

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Abstract

This project develops and evaluates an autonomous-navigation workflow for a physical TurtleBot using ROS and Nav2. A standard Nav2 planner provides the system baseline, while a custom RRT* global planner exposes the planning logic needed to study sampling, collision checking, rewiring, and path publication. The implementation connects map-aware planning to RViz inspection and real-robot execution. This public brief describes the system and demonstrated functionality without reproducing the original course report.

System design

The custom planner operates on the navigation cost map and builds a collision-aware search tree from the current robot pose toward a requested goal. The main components are:

- **Sampling and expansion:** goal-biased sampling and fixed-step expansion balance broad exploration with progress toward the target.
- **Map-aware collision checks:** KD-tree distance queries and edge sampling account for occupied cells, map resolution, robot footprint, and configurable obstacle inflation.
- **RRT* refinement:** best-parent selection and local rewiring reduce accumulated path cost as the tree grows.
- **ROS integration:** the selected route is backtracked, interpolated, converted to a ROS path, published to the global-path topic, and inspected in RViz before execution.

Validation

The navigation workflow was tested in both screen-recorded planning sessions and physical-robot trials. The public video intentionally presents the sequence in that order of evidence: a Nav2 physical run is followed by its corresponding screen recording, then the clip loops. The project demonstrates an end-to-end connection between a standard navigation baseline, an inspectable custom planner, and real-platform execution.